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COURSE NAME : NLP

Q1) **Section 1: Regular Expressions – Resume Information Extraction**

main.py Visualize Run

```
import re

resume = """
Name: Vijayalakshmi S
Email: vijay@gmail.com
Mobile: 9876543210
Skills: Python, SQL, Machine Learning, NLP
Experience: 3 years
"""

name = re.search(r"Name:\s*(.+)", resume)
email = re.search(r"[\w.-]+@[\w.-]+\.\w+", resume)
mobile = re.search(r"\b[6-9]\d{9}\b", resume)
experience = re.search(r"Experience:\s*(\d+)\s*years?", resume, re.I)

skills = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
found_skills = [
    skill for skill in skills
    if re.search(r"\b" + re.escape(skill) + r"\b",
                 resume, re.I)
]

years = int(experience.group(1)) if experience else 0
```

```
years = int(experience.group(1)) if experience else 0

print("RESUME SUMMARY")
print("Name:", name.group(1) if name else "Not found")
print("Email:", email.group() if email else "Not found")
print("Mobile:", mobile.group() if mobile else "Not found")
print("Skills:", ", ".join(found_skills))
print("Experience:", years, "years")

if years >= 2 and "Python" in found_skills:
    print("Eligibility: Eligible for shortlisting")
else:
    print("Eligibility: Not eligible")
```

Output

```
RESUME SUMMARY
Name: Vijayalakshmi S
Email: vijay@gmail.com
Mobile: 9876543210
Skills: Python, SQL, Machine Learning, NLP
Experience: 3 years
Eligibility: Eligible for shortlisting
```

Q2) **Section 2: Regular Expressions – Product Search System**

main.py

Visualize Run

```python

import re

products = [
    "Python Programming Book",
    "Java Programming Book",
    "Python Laptop",
    "Wireless Mouse",
    "Laptop Bag",
    "Smartphone",
    "Python Keyboard"
]

keyword = input("Enter search keyword: ")
print("1. Exact  2. Prefix  3. Suffix")
print("4. Partial 5. Case-insensitive")
choice = input("Enter choice: ")

if choice == "1":
    pattern = r"^" + re.escape(keyword) + r"$"
elif choice == "2":
    pattern = r"^" + re.escape(keyword)
elif choice == "3":
    pattern = re.escape(keyword) + r"$"
elif choice == "4":
    pattern = re.escape(keyword)
```

main.py

Visualize Run ▶

```
    pattern = r"^" + re.escape(keyword)
elif choice == "3":
    pattern = re.escape(keyword) + r"$"
elif choice == "4":
    pattern = re.escape(keyword)
elif choice == "5":
    pattern = re.escape(keyword)
else:
    print("Invalid choice")
    raise SystemExit

flags = re.I if choice == "5" else 0

matches = [
    product for product in products
    if re.search(pattern, product, flags)
]

print("\nMatching Products:")
if matches:
    for product in matches:
        print(product)
else:
    print("No products found")

print("Total matching products:", len(matches))
```

Output

```
Enter search keyword: python
1. Exact  2. Prefix  3. Suffix
4. Partial 5. Case-insensitive
Enter choice: 4

Matching Products:
No products found
Total matching products: 0
```

Q3) **Section 3: Regular Expressions – University Registration System**

main.py

Visualize

Run

```
import re

reg_no = input("Enter register number (e.g., 2312345678): ")
email = input("Enter institutional email: ")
course = input("Enter course code (e.g., CSE101): ")
semester = input("Enter semester (1-8): ")
mobile = input("Enter mobile number: ")

checks = {
    "Register Number": bool(re.fullmatch(r"\d{10}", reg_no)),
    "Institutional Email": bool(re.fullmatch(
        r"[\w.-]+@saveetha\.com", email, re.I)),
    "Course Code": bool(re.fullmatch(
        r"[A-Z]{3}\d{3}", course, re.I)),
    "Semester": bool(re.fullmatch(
        r"[1-8]", semester)),
    "Mobile Number": bool(re.fullmatch(
        r"[6-9]\d{9}", mobile))
}

print("\nVALIDATION REPORT")

for field, valid in checks.items():
    if valid:
        print(field + ": Valid")
```

```
print("\nVALIDATION REPORT")

for field, valid in checks.items():
    if valid:
        print(field + ": Valid")
    else:
        print(field + ": Invalid")

if all(checks.values()):
    print("\nRegistration Status: Successful")
else:
    print("\nRegistration Status: Failed")
```

Output

```
Enter register number (e.g., 2312345678): 2312345678
Enter institutional email: student@saveetha.com
Enter course code (e.g., CSE101): CSE101
Enter semester (1-8): 3
Enter mobile number: 9876543210

VALIDATION REPORT
Register Number: Valid
Institutional Email: Valid
Course Code: Valid
Semester: Valid
Mobile Number: Valid

Registration Status: Successful
```